NETRAX · Claude Code prompt pack

Clean dashboard, icon alerts, one alert per violation, voice, fire call, reliable Telegram / e-mail · Team ZENTRIX · 12 Nov 2026

1 · What we have to build

1. **Dashboard polishing first:** all pages in the same clean style, slightly bigger fonts, clear categories, equal panel sizes and padding, no secondary writing, every feature kept.
2. **Icon-based violations** with the worker ID and name; “Unregistered person” when unknown; ID, name and UNAUTHORISED for access violations.
3. **One alert per violation**, and the laptop alarm telling all violations with the right name.
4. **Fire / smoke call at once**, repeated until someone acknowledges or resolves it.
5. **Telegram and e-mail** that do not fail or get lost.
6. The control room **guided by the supervisor** (acknowledge, resolve, escalate).

Removed from the earlier “Judges’ prerequisites” plan

- **WhatsApp** channel: dropped, as decided.
- A **separate Supervisor page**: dropped. Instead every existing page is cleaned in the same style, so all features stay where the team knows them.
- **Icons on the video overlay**: not now. The video is the AI view; icons go on cards, entrance results and messages.
- Telegram Acknowledge buttons are a small optional extra in Prompt 10, only if quick.

2 · What we found in our own project

Area	What we measured in our project (12 Nov)	What it means
Dashboard pages	7 pages (control room, entrance, hardware, intelligence, admin, ROI, login), 4,653 lines. Admin alone has about 30 panels on one page. 167 font-size rules are between 10 and 13 px. Each page has 11–35 hint / sub-title texts.	The “messy” feeling: too many small texts and too many panels at the same level. Needs one style, categories, bigger fonts.
Violation display	Text chips “✖ NO HELMET”, “Missing PPE: GLOVES” in index.html, alerts.js and the incident drawer. The event stores only items_missing.	Needs icons, and the event must also carry worn / cannot-see items.
Alerts per violation	967 PPE events but only 333 different person + camera + missing-items combinations.	About 3 alerts for one violation. Fix: one open violation per person.
E-mail	913 sent, 203 failed: 124 × Gmail “daily sending limit exceeded”, 75 × no internet (DNS), 4 × timeout.	Too many mails (the duplicates), and no waiting for the internet to return.
Telegram	1,768 delivered, 128 failed: 115 × connection error, 10 × timeout. Some rows stuck in queued / sending.	Failures are internet drops, so retry longer and resend after reconnect.
Fire call	4 delivered, 15 failed (busy, no-answer, network). One call per fire; called again only after 120 s for Level 3 or escalation.	Must call at once and keep calling until someone acknowledges.
Laptop voice	Says “Attention worker W 0 0 4, Ritika … Please wear your helmet now”; unregistered = “unregistered person in the zone”. Browser voice can also speak.	Mostly right; make the sentence fixed, name first, all items, once, one voice only.

3 · How to use this pack

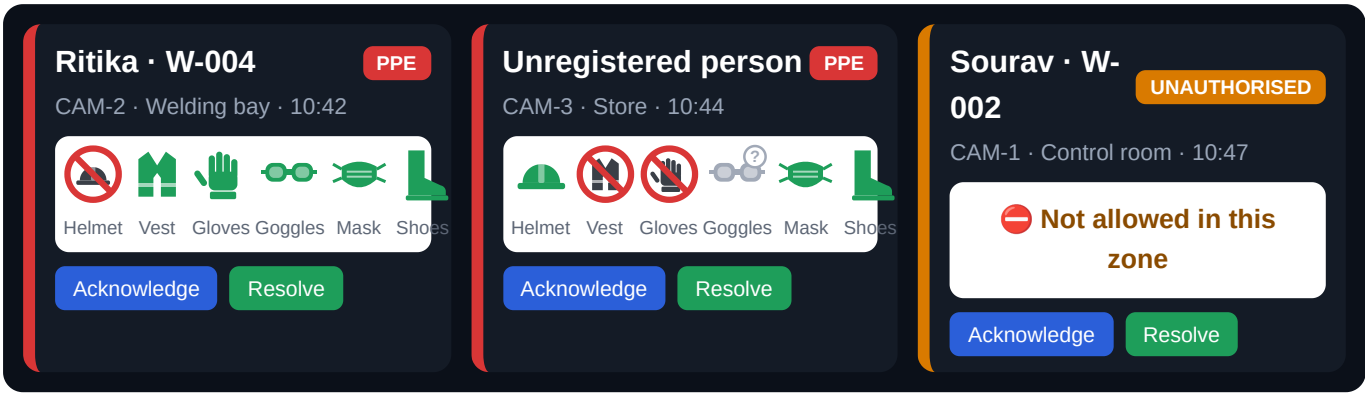
1. Open Claude Code in `C:\Users\Satyabrata\Desktop\sentineleye`.
2. Paste **one prompt at a time**, in order. Prompt 0 changes nothing; it only reads, backs up and takes screenshots.
3. After each prompt, do the **check** under it. If something broke, say “undo the last step from _to_delete/before_polish_12nov” before going on.
4. If time is short, the order already puts the dashboard first (Prompts 1–5), then alerts (6–10). Stop after any prompt and the system still works.
5. Total: about 7 h of work in total (Prompts 0–11). Steps 2–4 can be done by two people at the same time on different pages only after Step 1 is finished.

4 · Design target

Icons: 6 items × 3 states

Item	Worn (green)	Missing (red prohibition)	Cannot see (grey ?)
Helmet `helmet`			
Safety vest `vest`			
Gloves `gloves`			
Goggles `goggles`			
Mask `mask`			
Safety shoes `boots`			

Alert cards



Target alert cards (mock-up): registered worker · unregistered person · unauthorised entry. No sentences, only the person, place, time, icons and two buttons.

Rules every page follows

Rule	Value
Fonts	Body 16 px · small 14 px · time stamps 13 px (minimum) · panel titles 19 px · big numbers 34 px
Spacing	4 / 8 / 12 / 16 / 24 px only; panel padding 16 px; 16 px between panels; radius 12 px
Colours	Green = compliant / normal · Red = violation / fire · Amber = unauthorised / warning · Grey = cannot see / offline. Nothing else is coloured.
Panel	Icon + title + count + at most 2 buttons. No sub-title, no explanation (an (i) tooltip if really needed). Equal height in a row; long lists scroll inside.
Alert card	Person · camera · zone · time · icons · Acknowledge · Resolve. Everything else in the detail drawer.
Person	Registered “Name · W-004” · unknown “Unregistered person” · access “Name · W-002 + UNAUTHORISED”

5 · The prompts

Prompt 0 · Read, back up, screenshot (no changes)

15 min

```
You are working on NETRAX (folder: sentineleye), our edge-AI PPE / fire / smoke safety system
for the BPUT Hackathon final.
First read CLAUDE.md, then dashboard/ (index, entrance, hardware, intelligence, admin, roi,
login + nx.css, theme.css, alerts.js, zentrix.js, video.js), backend/main.py, backend/notify.py,
backend/actuators.py, config/notify.yaml.

Over the next prompts we will (1) clean and polish ALL dashboard pages in one common style, (2)
show PPE violations as ICONS with the worker ID + name, (3) send exactly ONE alert per
violation, (4) fix the laptop voice alarm text, (5) make fire / smoke calls immediate and
repeating until acknowledged, (6) make Telegram and e-mail reliable, (7) let the supervisor
guide every alert from the control room.

Rules for the whole job (repeat them to yourself before every change):
- Before changing any file, copy it to _to_delete/before_polish_12nov/<same path>.
- Do NOT touch models/, weights, training code, edge/engine_proc.py detection logic or
thresholds. No retraining.
- Never commit or print config/secrets.local.yaml, config/users.local.yaml, firmware/*/secrets.h
or data/.
- Keep EVERY existing feature, button, API route and WebSocket working. We remove clutter, not
functions.
- Only measured numbers may appear on screen. Do not invent data.
- After each step run: python -m edge.selftest  (all checks must still pass) and open the
changed page in the browser.

Now do only this:
1. Make the backup folder.
2. Start the system (run_all.bat or the usual way) and take a full-page screenshot of every
dashboard page at 1920x1080 and 1366x768 into _to_delete/before_polish_12nov/shots/ (Playwright
is fine).
3. Write me a short list per page: panels it has, small texts / hints / helper sentences it
shows, fonts below 14 px, and places where violations are written as text ("NO HELMET", "Missing
PPE: ...").
Do not change any code in this prompt.
```

Check before the next prompt:

- ☐ Backups and before-screenshots exist
- ☐ You got the per-page clutter list

Prompt 1 · Common design system + icon + person helpers

40 min

```
Step 1 - one common design system for all pages. Edit dashboard/nx.css (and theme.css only if
needed); keep Day / Night themes.

1. Type scale as CSS variables, used everywhere instead of fixed px:
   --fs-xs 13px (only time stamps), --fs-s 14px, --fs-m 16px (body, default), --fs-l 19px (panel
titles), --fs-xl 24px, --fs-num 34px (big numbers).
   Nothing on any page may be smaller than 13px. Today there are ~167 font-size rules at 10-
13px; replace them with the variables.
2. Spacing scale: --sp-1 4px, --sp-2 8px, --sp-3 12px, --sp-4 16px, --sp-5 24px. Panels: padding
var(--sp-4), gap between panels var(--sp-4), radius 12px, one border style, one shadow.
3. One panel component used by every page: .nx-panel > .nx-panel-head (icon + title + optional
count chip + optional 1-2 action buttons on the right) > .nx-panel-body. No subtitle line under
titles.
4. Status colours only from variables: --ok (green) compliant / normal, --crit (red) violation /
```

```
fire, --warn (amber) unauthorised / warning, --unk (grey) cannot see / offline. Green and red
are never used for decoration.
5. Grid helpers: .nx-grid-2 / -3 / -4 that fall back to 1 column under 1100px. Panels in the
same row get the same height (align-items: stretch), long lists scroll inside the panel body
(max-height) instead of stretching the page.
6. Buttons: one size (.btn 36px high, 14px text), .btn-primary, .btn-ok, .btn-danger, .btn-
ghost.
7. Make a new file dashboard/ppe_icons.js with ppeIcon(item, state, size) -> inline SVG string.
   items: helmet, vest, gloves, goggles, mask, boots.  states: "ok" = item drawn in green;
"miss" = item in dark grey inside a red prohibition circle with a red diagonal bar; "unk" = item
in grey with a small "?" badge.
   Simple flat original shapes (no copied artwork). Also export ppeRow(items_required, items_ok,
items_missing, items_uncertain, size) that returns the icons in a fixed order with a tiny label
under each and a title tooltip ("Helmet: missing").
8. Make dashboard/people.js with personLabel(e):
   registered worker -> "Name · W-004";  unregistered -> "Unregistered person" (+ " #17" track
id small);  unauthorised_entry -> "Name · W-002" plus an amber UNAUTHORISED badge.
   Hidden names (W-***) keep the existing privacy behaviour.
Include the two new scripts in every page that shows alerts. Do not restyle individual pages
yet; show me one test page (dashboard/_style_test.html, delete it at the end) with all
components, all icons in all 3 states, in Day and Night.
```

Check before the next prompt:

- ☐ Test page shows all components and 18 icon states in Day and Night
- ☐ selftest passes

Prompt 2 · Control room clean-up

50 min

```
Step 2 - clean the Control room (dashboard/index.html) using only the new components.

Keep all its features: live camera tiles, entrance alerts, zones & people, safety station,
system health, fire & smoke, PPE & access, incident details, mute, voice, Zentrix button, theme
switch.
Make it calm and readable from 2 m:
1. Top bar: logo, site name, clock, 4 KPI tiles only (Workers in view · Compliant · Open PPE
violations · Fire / smoke status). Big number + one word label. Remove other chips from the top
bar (move them into System health).
2. Layout, categorised: left 2/3 = Live cameras (grid of tiles, camera name + LIVE / OFFLINE dot
+ fps small in corner). Right 1/3 = "Alerts" panel (one list, newest first, fire always on top,
then PPE, then access).
   Below: one row with 3 equal panels: Zones & people · Safety station · System health.
   Fire & smoke panel only becomes big when there is a fire; otherwise it is one green line
"Fire / smoke: clear".
3. Remove helper sentences, explanations, hints, "how it works" texts, long sub-titles, repeated
labels, debug numbers. If something must stay for the jury, move it into a small (i) tooltip.
4. Alert cards = the icon card: personLabel(e), camera · zone · time, ppeRow(...) icons, and
only two buttons (Acknowledge, Resolve). Details (snapshot, confidence, event id, notification
log) open in the existing detail drawer, not on the card.
   No text like "NO HELMET" / "Missing PPE:" / "Required PPE missing" anywhere on the card.
5. When an alert's missing items change, update the icons on the same card (keep the existing
"fresh" highlight), do not add a second card.
Show me screenshots at 1920 and 1366, Day and Night, and the list of everything you removed or
moved.
```

Check before the next prompt:

- ☐ 4 KPI tiles, cameras left, one Alerts list right
- ☐ No 'NO HELMET' text; icons only
- ☐ All buttons still work

Prompt 3 · Entrance + Hardware pages

40 min

```
Step 3 - same clean style for Entrance (entrance.html) and Hardware (hardware.html).

Entrance: keep camera, entrance alerts, required PPE, devices, inside-the-site list, entrance
log.
- Result block = big PERMITTED (green) / NOT PERMITTED (red) + personLabel + ppeRow icons of the
checked items (use items_ok / items_missing / items_uncertain that entrance_checks already
stores). No sentences.
- "How verification works" panel: remove from the page, keep as an (i) tooltip on the panel
title.
- Required PPE panel: icons only (green = required here), one row per zone.
- Entrance log: table with time · person · result icon · items icons; 14px text.
Hardware: keep live sensors, fire & smoke cross-check, in the area now, PPE contact sensors,
live location, RFID attendance.
- Sensor tiles all the same size: icon, name, big value + unit, status dot (green / amber /
red). No paragraphs under them.
- Group into 3 categories with headers: Environment (temperature, humidity, gas, dust, flame) ·
Safety (fire cross-check, PPE contact, tilt / fall) · People & location (RFID, in the area,
GPS).
Same component, spacing and font rules as the control room. Screenshots again.
```

Check before the next prompt:

- ☐ PERMITTED / NOT PERMITTED + icons
- ☐ Sensor tiles equal size, 3 groups

Prompt 4 · Intelligence, Admin (tabs), ROI, Login

50 min

```
Step 4 - Intelligence (intelligence.html), Admin (admin.html), ROI (roi.html) and Login
(login.html).

Admin today has ~30 panels on one long page. Keep every one of them, but categorise into tabs at
the top (one page, tab bar, remember the last tab in localStorage with try/catch):
  Overview (violations last 7 days, most missed PPE, by zone, zone x hour, live activity, shift
summary, compliance mix)
  Incidents & reports (search history, generate a report, summary / by zone / camera / worker,
records)
  Workers (register a worker, registered workers)
  Alerts & delivery (send a test message, delivery log)
  Site setup (cameras, zones / required PPE / access, speaker & siren, safety station)
  System (dashboard users, audit log, components, measured latency, live end-to-end run)
Charts: same colours as status variables; text in charts >= 13px; most-missed-PPE chart uses the
ppe icons as row labels.
Intelligence: same tab idea for its sections; remove explanatory sentences; same tile sizes.
ROI and Login: apply the new panel, buttons and fonts only.
Make all pages share one header / navigation bar (same height, same order of links, active page
highlighted).
Screenshots of every tab.
```

Check before the next prompt:

- ☐ Admin in 6 tabs, nothing lost
- ☐ Same header on every page

Prompt 5 · Icons + correct person in every alert (data)

35 min

```
Step 5 - icons + correct person in EVERY alert (data side).
```

```
1. Today a ppe_missing event only carries items_missing. In edge/state_machine.py (where the
event payload is built - do not change how the decision is made) also send items_ok,
items_uncertain and items_required from the per-item states the tracker already has.
2. backend/db.py: add the columns items_ok, items_uncertain, items_required to events the safe
way (ALTER TABLE if missing, like the notifications table). backend/main.py stores and returns
them; old events without them must still display (treat as unknown).
3. Use ppeRow everywhere a PPE alert is shown: control room, alerts.js pop-ups on all pages,
incident drawer, admin incident search, entrance log.
4. Video overlay (edge/annotate.py _item_column): keep it as it is (text) unless I ask; it is
the AI view.
5. Identity rules, the same on screen, Telegram, e-mail and voice:
   registered worker  -> name + worker ID + icons
   unregistered       -> "Unregistered person" + icons (no fake name)
   unauthorised entry -> name + worker ID + "UNAUTHORISED" (amber) + zone
   Check worker_name is filled from the workers table whenever worker_id is set; fix the places
where an alert shows an ID without the name.
Add selftest checks for: payload has the 3 new fields; old event without them renders;
personLabel for the 3 cases.
```

Check before the next prompt:

- ☐ New events carry items_ok / items_uncertain
- ☐ Old events still open
- ☐ 3 identity cases correct

Prompt 6 · One alert per violation

40 min

```
Step 6 - exactly ONE alert per violation.

Measured on our database: 967 ppe_missing events but only 333 different (person/track + camera +
missing items) combinations, so a violation produced ~3 alerts, 3 Telegram messages and 3 e-
mails. This also used up the Gmail daily limit.
Find why (track id changes when a person is lost for a moment, the same worker seen by 2
cameras, a resolved event re-opening, cooldown only per zone+type) and fix it with an "open
violation" rule:
- Key = worker_id if known, else the person track, + violation type (ppe_missing /
unauthorized_entry) + zone.
- While a violation with that key is open or acknowledged, a new detection UPDATES it (missing
items, last_seen, snapshot) instead of creating a new event. Added items keep the existing "Also
missing" update (one message, not a new alert).
- A new alert for the same key only after the supervisor resolved it AND the person is seen
violating again after a grace time (config: notify.yaml -> one_alert: {regrace_s: 120}).
- Fire / smoke are not merged with PPE and never delayed.
- Notifications: one row per event per channel (already), plus the update message is sent only
on Telegram.
Show me before/after counts using the replay or a live 2-minute test with one person removing
the helmet, walking out of view and back.
```

Check before the next prompt:

- ☐ Helmet-off test gives 1 event, 1 Telegram, 1 e-mail

Prompt 7 · Laptop voice alarm with names

20 min

```
Step 7 - laptop voice alarm (backend/main.py ppe_text / access_text / ok_text,
backend/actuators.py Speaker).

The voice must say ALL violations of the person with the correct name, once per alert, in a
fixed clear sentence:
- registered:   "Attention Ritika, worker W 0 0 4. Helmet and gloves missing. Please wear them
```

```
now."
- unregistered: "Attention, unregistered person in the Welding bay. Helmet missing. Please wear
it now. Report to the supervisor."
- unauthorised: "Attention Sourav, worker W 0 0 2. You are not authorised in the Control room.
Please leave now."
- fire:         unchanged siren + fire message, always first.
Rules: speak only when the alert is new or an item was added (then: "Ritika, gloves also
missing."), never repeat the same sentence again for the same open violation unless the reminder
timer in config says so; never speak a name we do not know; items in the same order as the
icons; use item_names from config.
The dashboard voice (zentrix.js speechSynthesis) must not speak the same alert at the same time
as the laptop speaker: keep only one of them for alerts (the laptop SAPI speaker), the dashboard
voice stays for the Zentrix assistant.
Add selftest checks for the 3 sentences.
```

Check before the next prompt:

- ☐ 3 sentences correct, said once
- ☐ No double voice from the browser

Prompt 8 · Immediate, repeating fire call

30 min

```
Step 8 - fire / smoke phone call: IMMEDIATE and REPEATED until someone acts.

Today: config/notify.yaml phone.recall_after_s = 120 and backend/notify.py makes one call per
fire and calls again only for Level 3 or an escalation. Our delivery log shows 15 failed calls,
Twilio "busy" and "no-answer" and network errors - those were not retried quickly.
Change to:
1. The FIRST fire or smoke detection (any level, camera or safety station) queues the call at
once - no waiting for level 2 or for the escalation timer. Telegram goes out at the same moment.
2. Repeat the call every phone.repeat_every_s (default 60 s) while the event is open or
escalated. Stop immediately when a supervisor presses Acknowledge or Resolve on the dashboard
(or the event is closed). Max phone.max_calls (default 10) per event.
3. busy / no-answer / failed / network error -> next try after 20 s, not 60 s.
4. Every call attempt is a row in notifications (attempt number, provider status) so the
Delivery log shows it.
5. Call text: "NETRAX fire alarm. Fire detected in <zone> by <camera / station> at <time>. Open
the dashboard and acknowledge." read twice.
6. PPE / access alerts still never call (only_fire_smoke stays true).
Test with a test fire event: the call starts within 5 s, repeats, stops within one cycle after
Acknowledge. Show the log.
```

Check before the next prompt:

- ☐ Call within 5 s
- ☐ Repeats every 60 s
- ☐ Stops after Acknowledge

Prompt 9 · Reliable Telegram and e-mail

45 min

```
Step 9 - Telegram and e-mail must not fail.

Measured from our notifications table:
- e-mail: 913 sent, 203 failed -> 124 x Gmail "550 5.4.5 Daily user sending limit exceeded", 75
x "getaddrinfo failed" (no internet), 4 x SMTP timeout.
- Telegram: 1768 delivered, 128 failed -> 115 x network ConnectionError, 10 x ReadTimeout.
- some rows stuck in "queued" / "sending".
Fix:
1. Fewer e-mails: thanks to Step 6 there is one alert per violation. E-mail only for: fire /
smoke, Critical / High PPE, unauthorised entry, escalations. Everything else goes in one e-mail
```

```
digest every 15 min (config). Add a daily e-mail counter with a safe limit (config
email.daily_max, default 300) and when it is reached send only fire mails and tell the
dashboard.
2. Offline-safe: if there is no internet (DNS / connection error), do NOT count it as a failed
attempt: keep the message in the outbox and retry with backoff (5 s, 15 s, 30 s, then every 60
s) for up to 2 hours; send in order when the internet returns. Messages older than that are
marked "expired" with the reason.
3. On start-up, rows left in "queued" / "sending" from a crash are picked up again (not
forgotten).
4. Telegram: send the snapshot photo + short caption with the person and missing items as text
emoji lines (Helmet ❌, Vest ✅, Goggles ❓), one message per alert; updates as a reply to that
first message.
5. SMTP: reuse one connection, 20 s timeout, retry once on SMTPServerDisconnected.
6. Delivery log on the Admin page shows per channel: sent / waiting for internet / failed
(reason in plain words).
Tests: unplug Wi-Fi for 2 minutes during a test alert -> messages arrive after reconnecting,
none lost, none duplicated.
```

Check before the next prompt:

- ☐ Wi-Fi-off test: nothing lost, nothing doubled
- ☐ E-mail count under the daily limit

Prompt 10 · Supervisor-guided control room

30 min

```
Step 10 - the control room is guided by the supervisor.

- Every alert stays on the control room until a supervisor acknowledges and then resolves it
(viewer role can only look; buttons hidden for viewers, API still checks the role).
- Acknowledge = "I have seen it" -> stops the repeating call and the escalation timer, card
turns from red-blinking to red-steady, shows "Ack by <user> <time>".
- Resolve = "fixed" -> optional one-tap reason (PPE worn now / false alarm / worker left / fire
out), card leaves the list, goes to history.
- Not acknowledged in escalation time (config) -> escalation exactly as today.
- Acknowledge and resolve also possible from the Telegram message: inline buttons "Acknowledge"
/ "Resolve" (Telegram callback with the event id; check the chat id). Only if this is less than
30 min; otherwise skip and tell me.
- Every action is written to the audit log (already exists) and shown in the incident drawer.
```

Check before the next prompt:

- ☐ Viewer cannot act
- ☐ Ack stops the call
- ☐ Actions in audit log

Prompt 11 · Final polish, test, commit

30 min

```
Step 11 - final polish and test.

1. Go through every page and tab at 1920x1080, 1366x768 and a phone width (390 px), Day and
Night. Fix: overlapping text, cut-off numbers, panels of different heights in one row, scroll
bars inside small boxes, any font < 13 px, any leftover sentence or "NO <ITEM>" text.
2. Run python -m edge.selftest (all pass) and a 5-minute live run with 3 cameras: helmet off ->
one red icon card + one Telegram + one voice sentence with the name; helmet on -> resolve; test
fire -> immediate repeated call, stops on acknowledge.
3. Delete dashboard/_style_test.html. Make sure nothing in _to_delete/ is needed by the app.
4. Give me a before/after screenshot pair for every page into docs/polish_12nov/, and a short
change list (file -> what changed). Update CLAUDE.md with the new design rules and the new
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config keys.
5. Commit locally with a clear message (never push secrets / data/). Do not push.
```

Check before the next prompt:

☐ Before/after screenshots saved

☐ selftest passes

☐ Committed locally, not pushed

6 · After the last prompt: demo check

☐ Every page opens with the same header, big readable numbers, no sentences.

☐ Helmet off: one red card with the red helmet icon, the name and ID; one Telegram with photo; the laptop says the name and the item once.

☐ Unknown person: “Unregistered person” card with icons.

☐ Worker in a forbidden zone: amber UNAUTHORISED card with name and ID.

☐ Test fire: phone rings within 5 s, again every 60 s, stops after Acknowledge.

☐ Wi-Fi off for 2 min: alerts arrive after reconnect, none lost, none doubled.

☐ Viewer login cannot acknowledge or resolve.